Day 18 - 16th july 2025

Pending:

Heap Sort
Radix sort

Task01

What kind of collision resolution strategy is implemented in the below Hash Table ?
import java.util.*;

```java
import java.util.*;

class Task01 {

    LinkedList<Entry>[] data = new LinkedList[10]; // Creating a hash table with 10 buckets

    // Method to insert key-value pairs into the hash table
    public void put(String keyval, int value) {  4 usages
        int index = Math.abs(keyval.hashCode() % data.length);  // Compute hash index

        if (data[index] == null) {
            data[index] = new LinkedList<>();  // If no linked list exists, create one
        }

        // Check if the key already exists and update the value if so
        for (Entry e : data[index]) {
            if (e.keyval.equals(keyval)) {
                e.value = value;  // Update the value if key is found
                return;
            }
        }

        // Add the new entry to the list
        data[index].add(new Entry(keyval, value));
    }

    // Method to retrieve the value for a given key
```

```java
public int get(String keyval) {  3 usages

    if (data[index] != null) {
        for (Entry e : data[index]) {
            if (e.keyval.equals(keyval)) {
                return e.value;  // Return the value if the key is found
            }
        }
    }
    return -1;  // Return -1 if the key doesn't exist
}

// Method to display the contents of the hash table
public void display() {  1 usage
    for (int i = 0; i < data.length; i++) {
        if (data[i] != null) {
            System.out.print("Bucket " + i + ": ");
            for (Entry e : data[i]) {
                System.out.print("[" + e.keyval + "=" + e.value + "] ");
            }
            System.out.println();
        }
    }
}
```

```
            System.out.print("Bucket " + i + ": ");
            for (Entry e : data[i]) {
                System.out.print("[" + e.keyval + "=" + e.value + "] ");
            }
            System.out.println();
        }
    }
}

// Entry class representing a key-value pair
static class Entry {  5 usages
    String keyval;  4 usages
    int value;  4 usages

    Entry(String k, int v) {  1 usage
        keyval = k;
        value = v;
    }
}

public static void main(String[] args) {
    Task01 hashTable = new Task01();

    // Adding some key-value pairs
```

Task 02:
Wap to take input from the user a 5 digit no and display digit by digit in the output

Hint:
If input is  456897
Output:
units digit is 7
Ones digit is 9
Hundreds digit is 8
Thousands digit is 6
10 thousands digit is 5
Lakhs digit is 4

10.14 to 10.18

```java
Rename usages
public class DigitByDigit1 {
    public static void main(String[] args) {
        // Create a scanner object to take input from the user
        Scanner scanner = new Scanner(System.in);

        // Ask the user for a number
        System.out.print("Enter a number: ");
        int number = scanner.nextInt();

        // Step 1: Display the digits one by one
        // Extract and display each digit from right to left
        int unitsDigit = number % 10;               // Unit place
        int tensDigit = (number / 10) % 10;          // Tens place
        int hundredsDigit = (number / 100) % 10;     // Hundreds place
        int thousandsDigit = (number / 1000) % 10;   // Thousands place
        int tenThousandsDigit = (number / 10000) % 10; // Ten-thousands place

        // Display the digits (Works for a number with at most 5 digits)
        System.out.println("Units digit is: " + unitsDigit);
        System.out.println("Tens digit is: " + tensDigit);
        System.out.println("Hundreds digit is: " + hundredsDigit);
        System.out.println("Thousands digit is: " + thousandsDigit);
        System.out.println("Ten-thousands digit is: " + tenThousandsDigit);
```

Task 03:
Wap to take number from the user and display the no of digit it has

HInt:
If input is:
10,000
Output:
Its a 5 digit number

```java
import java.util.Scanner;

Rename usages
public class DigitByDigit1 {
    public static void main(String[] args) {
        // Create a scanner object to take input from the user
        Scanner scanner = new Scanner(System.in);

        // Ask the user for a number
        System.out.print("Enter a number: ");
        int number = scanner.nextInt();

        // Step 1: Display the digits one by one
        // Extract and display each digit from right to left
        int unitsDigit = number % 10;               // Unit place
        int tensDigit = (number / 10) % 10;          // Tens place
        int hundredsDigit = (number / 100) % 10;     // Hundreds place
        int thousandsDigit = (number / 1000) % 10;   // Thousands place
        int tenThousandsDigit = (number / 10000) % 10; // Ten-thousands place

        // Display the digits (Works for a number with at most 5 digits)
        System.out.println("Units digit is: " + unitsDigit);
        System.out.println("Tens digit is: " + tensDigit);
        System.out.println("Hundreds digit is: " + hundredsDigit);
        System.out.println("Thousands digit is: " + thousandsDigit);
        System.out.println("Ten-thousands digit is: " + tenThousandsDigit);
```

```java
        int thousandsDigit = (number / 1000) % 10;   // Thousands place
        int tenThousandsDigit = (number / 10000) % 10; // Ten-thousands place

        // Display the digits (Works for a number with at most 5 digits)
        System.out.println("Units digit is: " + unitsDigit);
        System.out.println("Tens digit is: " + tensDigit);
        System.out.println("Hundreds digit is: " + hundredsDigit);
        System.out.println("Thousands digit is: " + thousandsDigit);
        System.out.println("Ten-thousands digit is: " + tenThousandsDigit);

        // Step 2: Display the number of digits
        // Calculate the number of digits in the number
        int digitCount = (int) Math.log10(number) + 1;

        // Display the number of digits
        System.out.println("This is a " + digitCount + " digit number.");
    }
}
```



```
C:\Program Files\Java\jdk-17\b
Enter a number: 12345
Units digit is: 5
Tens digit is: 4
Hundreds digit is: 3
Thousands digit is: 2
Ten-thousands digit is: 1
```

```
Thousands digit is: 2
Ten-thousands digit is: 1
This is a 5 digit number.

Process finished with exit code 0
```

```java
    public static void main(String[] args) {
        Task01 hashTable = new Task01();

        // Adding some key-value pairs
        hashTable.put( keyval: "Apple", value: 10);
        hashTable.put( keyval: "Banana", value: 20);
        hashTable.put( keyval: "Cherry", value: 30);
        hashTable.put( keyval: "Date", value: 40);

        // Displaying the hash table contents
        System.out.println("Hash Table Contents:");
        hashTable.display();

        // Retrieve value for a specific key
        System.out.println("\nValue for 'Banana': " + hashTable.get("Banana"));
        System.out.println("Value for 'Apple': " + hashTable.get("Apple"));
        System.out.println("Value for 'Orange': " + hashTable.get("Orange"));
    }
}
```

```
Value for 'Banana': 20
Value for 'Apple': 10
Value for 'Orange': -1

Process finished with exit code 0
```

Task 02:
Wap to take input from the user a 5 digit no and display digit by digit in the output

Hint:
If input is  456897
Output:
units digit is 7
Ones digit is 9
Hundreds digit is 8
Thousands digit is 6
10 thousands digit is 5

Lakhs digit is 4

```java
import java.util.Scanner;

public class DigitByDigit {
    public static void main(String[] args) {
        // Create a scanner object to take input from the user
        Scanner scanner = new Scanner(System.in);

        // Ask the user for a 5-digit number
        System.out.print("Enter a 5-digit number: ");
        int number = scanner.nextInt();

        // Validate if the input is a 5-digit number
        if (number < 10000 || number > 99999) {
            System.out.println("Please enter a valid 5-digit number.");
            return;
        }

        // Extract and display each digit from right to left
        int unitsDigit = number % 10;               // Unit place
        int tensDigit = (number / 10) % 10;          // Tens place
        int hundredsDigit = (number / 100) % 10;     // Hundreds place
        int thousandsDigit = (number / 1000) % 10;   // Thousands place
        int tenThousandsDigit = (number / 10000) % 10; // Ten-thousands place

        // Display the digits
        System.out.println("Units digit is: " + unitsDigit);
        System.out.println("Tens digit is: " + tensDigit);
```

```java
        // Extract and display each digit from right to left
        int unitsDigit = number % 10;               // Unit place
        int tensDigit = (number / 10) % 10;          // Tens place
        int hundredsDigit = (number / 100) % 10;     // Hundreds place
        int thousandsDigit = (number / 1000) % 10;   // Thousands place
        int tenThousandsDigit = (number / 10000) % 10; // Ten-thousands place

        // Display the digits
        System.out.println("Units digit is: " + unitsDigit);
        System.out.println("Tens digit is: " + tensDigit);
        System.out.println("Hundreds digit is: " + hundredsDigit);
        System.out.println("Thousands digit is: " + thousandsDigit);
        System.out.println("Ten-thousands digit is: " + tenThousandsDigit);
    }
}
```

```
Enter a 5-digit number: 12345
Units digit is: 5
Tens digit is: 4
Hundreds digit is: 3
Thousands digit is: 2
Ten-thousands digit is: 1
git > src > DigitByDigit
```

Task 04:

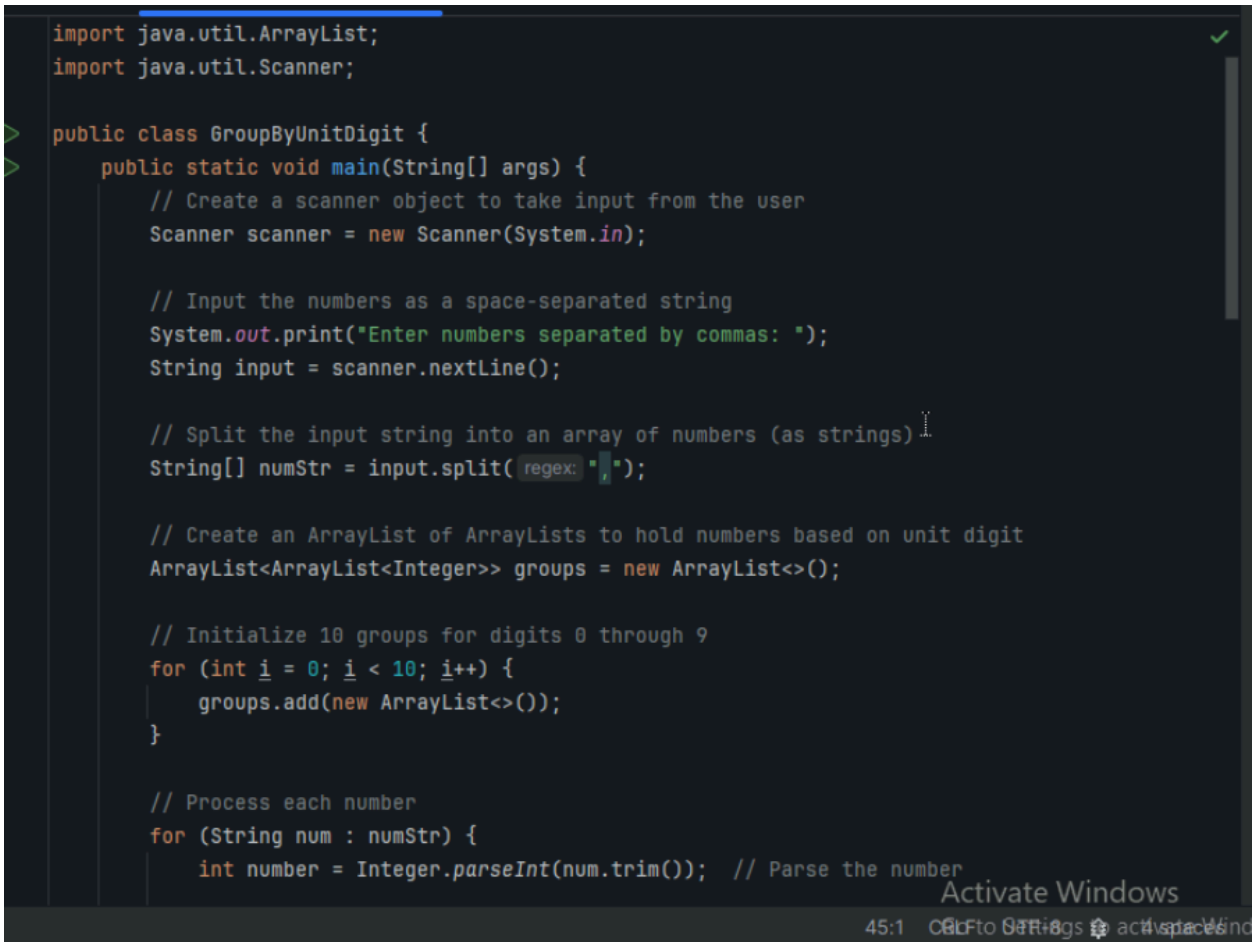
Wap to display the groups of digits depending upon the unit digits

Hint:
If input is 45,81, 85,100,20. 95,60,10,21
Output:
Array 1 has : 100,20,60,10

Array 2 has : 81, 21

Array 3 has : 45 , 85 ,95

10.41 to 10.45



```java
import java.util.ArrayList;
import java.util.Scanner;

public class GroupByUnitDigit {
    public static void main(String[] args) {
        // Create a scanner object to take input from the user
        Scanner scanner = new Scanner(System.in);

        // Input the numbers as a space-separated string
        System.out.print("Enter numbers separated by commas: ");
        String input = scanner.nextLine();

        // Split the input string into an array of numbers (as strings)
        String[] numStr = input.split( regex: ",");

        // Create an ArrayList of ArrayLists to hold numbers based on unit digit
        ArrayList<ArrayList<Integer>> groups = new ArrayList<>();

        // Initialize 10 groups for digits 0 through 9
        for (int i = 0; i < 10; i++) {
            groups.add(new ArrayList<>());
        }

        // Process each number
        for (String num : numStr) {
            int number = Integer.parseInt(num.trim());  // Parse the number
```

```
    public static void main(String[] args) {
        int number = Integer.parseInt(num.trim());  // Parse the number

        // Get the unit digit using modulo 10
        int unitDigit = number % 10;

        // Add the number to the correct group based on its unit digit
        groups.get(unitDigit).add(number);
    }

    // Display the groups with non-empty lists
    for (int i = 0; i < 10; i++) {
        if (!groups.get(i).isEmpty()) {
            System.out.println("Array " + (i + 1) + " has: " + groups.get(i));
        }
    }

    scanner.close(); // Close the scanner
}
}
```

Task 5
Algorithm for radix sort

## Step 1:

**Find the number with the most digits** in the list.
This tells us how many times we need to sort (once for each digit position).

## Step 2:

**Start with the digit in the units place** (rightmost digit).

## Step 3:

**Group the numbers based on the digit at that position.**
For example, group all numbers that have 0 in the unit place together, then all that have 1, then 2, and so on up to 9.

## Step 4:

**Rearrange the numbers according to those groups.**
Now the list is partially sorted based on the current digit.

## Step 5:

**Move to the next digit to the left** (tens place), and repeat steps 3 and 4.

## Step 6:

Keep repeating this process for the **hundreds place**, **thousands place**, etc., until you've sorted by all digit positions

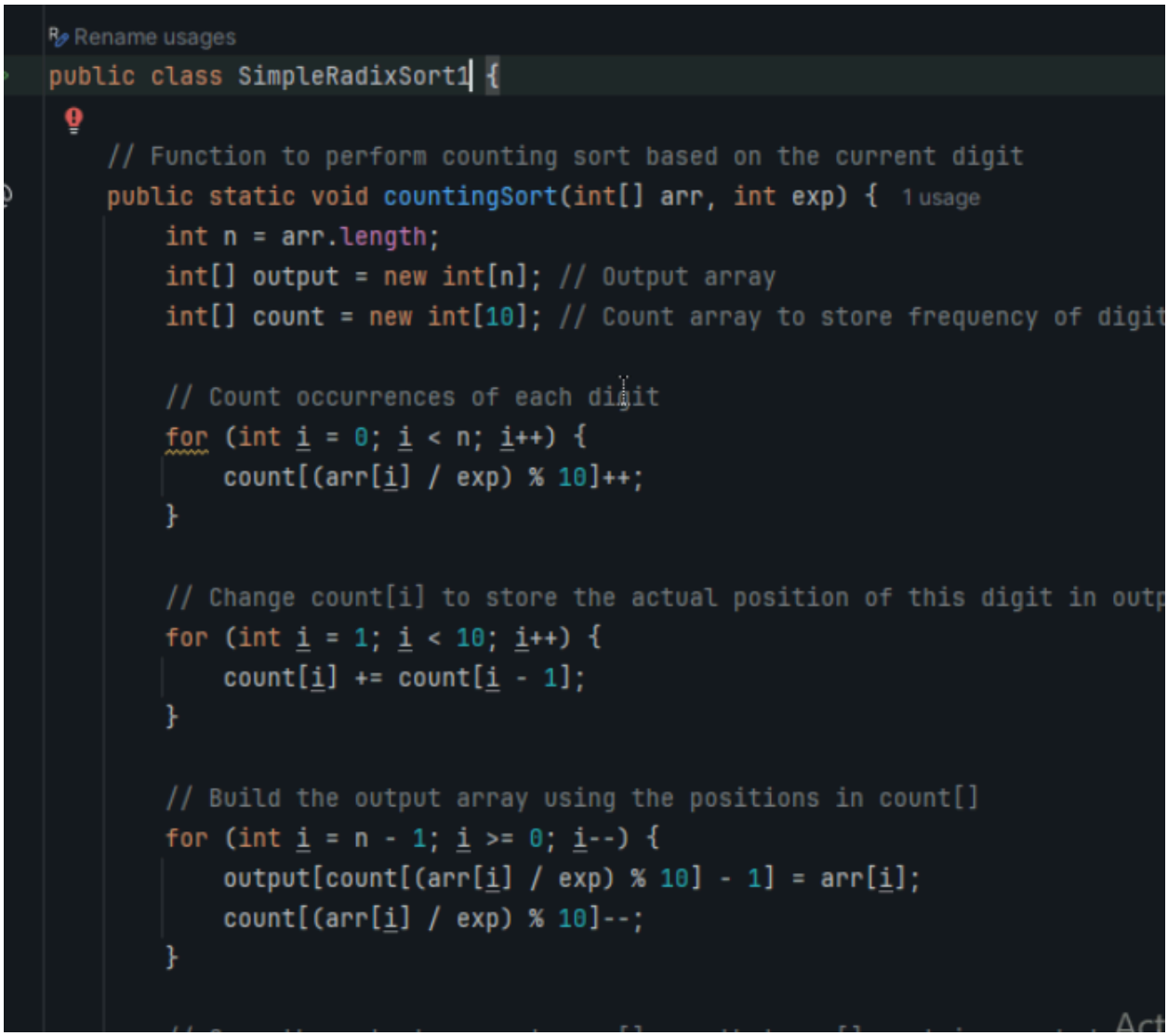
## Step 7:

After the final pass, the list will be **fully sorted**.

Task 6
Psudocode for Radix sort

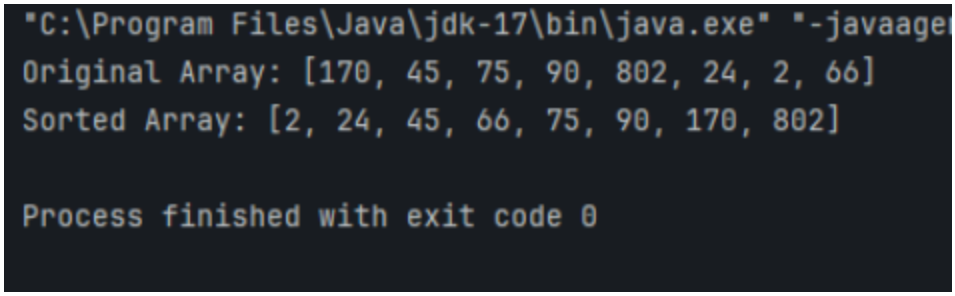
```
RadixSort(array):
    maxNumber = find the largest number in array
    digitPlace = 1

    while maxNumber / digitPlace > 0:
    sort array based on digit at digitPlace
    digitPlace = digitPlace * 10
```

Task 7
Full code


Rename usages
public class SimpleRadixSort1 {
// Function to perform counting sort based on the current digit
public static void countingSort(int[] arr, int exp) { 1 usage
int n = arr.length;
int[] output = new int[n]; // Output array
int[] count = new int[10]; // Count array to store frequency of digit
// Count occurrences of each digit
for (int i = 0; i < n; i++) {
count[(arr[i] / exp) % 10]++;
}
// Change count[i] to store the actual position of this digit in outp
for (int i = 1; i < 10; i++) {
count[i] += count[i - 1];
}
// Build the output array using the positions in count[]
for (int i = n - 1; i >= 0; i--) {
output[count[(arr[i] / exp) % 10] - 1] = arr[i];
count[(arr[i] / exp) % 10]--;
}

```
public static void countingSort(int[] arr, int exp) {  1 usage
    // Build the output array using the positions in count[]
    for (int i = n - 1; i >= 0; i--) {
        output[count[(arr[i] / exp) % 10] - 1] = arr[i];
        count[(arr[i] / exp) % 10]--;
    }

    // Copy the output array to arr[], so that arr[] contains sorted numbers
    System.arraycopy(output, srcPos: 0, arr, destPos: 0, n);
}

// Main function to implement Radix Sort
public static void radixSort(int[] arr) {  1 usage
    // Find the maximum number to determine the number of digits
    int max = Arrays.stream(arr).max().getAsInt();

    // Perform counting sort for every digit (units, tens, hundreds, etc.)
    for (int exp = 1; max / exp > 0; exp *= 10) {
        countingSort(arr, exp); // Sort based on each digit
```

```
public static void radixSort(int[] arr) {  1 usage
    // Find the maximum number to determine the number of digits
    int max = Arrays.stream(arr).max().getAsInt();

    // Perform counting sort for every digit (units, tens, hundreds, etc.)
    for (int exp = 1; max / exp > 0; exp *= 10) {
        countingSort(arr, exp); // Sort based on each digit
    }
}

public static void main(String[] args) {
    int[] arr = {170, 45, 75, 90, 802, 24, 2, 66}; // Example array

    System.out.println("Original Array: " + Arrays.toString(arr));

    // Apply Radix Sort
    radixSort(arr);

    System.out.println("Sorted Array: " + Arrays.toString(arr));
}
}
```




Task 08:

Do you find any significance change between the breadthFirstSearchRecursive() approach compared to the standard BFS?

Yes, there is a significant change between the two approaches.

> Standard Breadth-First Search (BFS) is defined as a graph traversal technique that uses a queue to explore nodes level by level, starting from a given source node. It is naturally implemented using an iterative approach.
>
> In contrast, `breadthFirstSearchRecursive()` is an unconventional approach that attempts to perform BFS using recursion, which relies on the call stack instead of explicit looping.

Standard BFS uses a queue data structure for level-order traversal.

Recursive BFS uses recursive function calls to simulate the same process.
Standard BFS is more efficient, widely accepted, and aligns with the definition of BFS.

Recursive BFS deviates from this standard definition and is more complex, less practical, and can cause performance issues due to deep recursion.

> Will it achieve the same result but emphasizes recursive style using the same level-order logic with explicit queue management?

Yes!

- Recursive BFS emulates the same level-order traversal.
- It still uses an explicit queue, just managed through recursive function calls.
- The main difference is the style of implementation (recursive vs iterative), but the logic remains the same.

This is the correct statement.

Task 09:

What is memoization?
Write in own words with examples.

Memoization is like keeping a cheat sheet for your program. When your program solves a small problem, it writes down the answer. Later, if it needs to solve the same small problem again, it just looks at the cheat sheet instead of solving it all over. This saves a lot of time because it doesn't repeat the same work.

For example, if you want to find the 5th number in the Fibonacci sequence, you normally have to find the 4th and 3rd numbers first. But finding the 3rd number might happen many times. With memoization, once you find the 3rd number, you save it. Next time you need it, you just use the saved answer.

So, memoization helps your program remember past answers to avoid doing extra work and run faster.

## Example:

Imagine you want to find the Fibonacci number for 5.

- Normally, you calculate Fibonacci(5) by calculating Fibonacci(4) and Fibonacci(3).
- Then to calculate Fibonacci(4), you calculate Fibonacci(3) and Fibonacci(2), and so on.
- But you see Fibonacci(3) gets calculated multiple times.

With memoization, once you calculate Fibonacci(3) once, you save it in memory. Next time you need Fibonacci(3), you just return the saved value instead of calculating again.

Task 10:
What do you understand by Dynamic Programming
Write in your own words with examples.
Dynamic programming is like solving a big problem by breaking it into smaller parts, solving each part only once, and then remembering those solutions to use them again whenever needed. Instead of repeating the same work over and over, you store answers to smaller problems and build up to the final answer efficiently.

It’s especially useful when a problem has overlapping subproblems — meaning many parts of the problem repeat themselves. By saving the answers to these repeated parts, you avoid wasting time.

Think of it like solving a puzzle step-by-step, remembering which pieces fit where, so you don’t try the same piece in the same place multiple times.

Imagine you want to climb a staircase with 5 steps, and you can take either 1 step or 2 steps at a time. To find out how many ways there are to reach the top, you could think:

- To get to step 5, you could come from step 4 or step 3.
- To get to step 4, you could come from step 3 or step 2.
- And so on...

Instead of calculating the ways to get to step 3 and 4 multiple times, dynamic programming helps you remember those answers so when you need them again, you just use the saved result. This makes the problem easier and faster to solve.

---

Task 11:

Can you write fibonacci series using dynamic programming?

```java
public class FibonacciDP {  no usages

    // Function to calculate Fibonacci series up to n using dynamic programming (bottom-up approa
    public static int fibonacci(int n) {
        // If n is 0 or 1, return n as the result (base cases)
        if (n == 0) return 0;
        if (n == 1) return 1;

        // Create an array to store Fibonacci numbers up to n
        int[] dp = new int[n + 1];

        // Base cases
        dp[0] = 0;
        dp[1] = 1;

        // Calculate Fibonacci numbers from 2 to n
        for (int i = 2; i <= n; i++) {
            dp[i] = dp[i - 1] + dp[i - 2];  // Fibonacci relation
        }

        return dp[n];  // Return the nth Fibonacci number
    }

    public static void main(String[] args) {
        int n = 10;  // Example value for n
        System.out.println("Fibonacci number at index " + n + " is: " + fibonacci(n));
    }
```

```java
public static int fibonacci(int n) {

    // Base cases
    dp[0] = 0;
    dp[1] = 1;

    // Calculate Fibonacci numbers from 2 to n
    for (int i = 2; i <= n; i++) {
        dp[i] = dp[i - 1] + dp[i - 2];  // Fibonacci relation
    }

    return dp[n];  // Return the nth Fibonacci number
}

public static void main(String[] args) {
    int n = 10;  // Example value for n
    System.out.println("Fibonacci number at index " + n + " is: " + fibonacci(n));
}
}
```

```
"C:\Program Files\Java\jdk-17\bin\java.ex
Fibonacci number at index 10 is: 55

Process finished with exit code 0
```

TSK 13:

how can you say recursive functions maintain the state of each call during execution?

1. Each recursive call creates a new thread, and context switching maintains state.

2. Recursive functions store state in global variables accessible across calls.

3. The system call stack tracks local variables and return addresses for each recursive invocation.

4. Recursive functions replicate the heap structure to keep values between calls.

Task 14:

Iterative implementations use less memory as they do not require stack frames for each call.
True

with regard to the difference in memory usage between recursive and iterative implementations of the same algorithm?
the above statement is true or false

True

Task 15:

Recursion that lacks a proper base case or makes too many nested calls, exhausting the call stack. – true

with regard to stack overflow error in recursive functions.. do you think above statement is true

Task 16:

what happens when inserting keys with the same hash in this custom hash map

```
public class HashCollision {

    static class Entry {
       String key;
       int value;

       Entry(String key, int value) {
          this.key = key;
          this.value = value;
       }
    }

    List<Entry>[] table = new AL[10];

    public void put(String key, int val) {
       int index = Math.abs(key.hashCode() % table.length);
```

```
        if (table[index] == null) {
            table[index] = new AL<>();
        }

        table[index].add(new Entry(key, val));
    }
}
```

1. Insertion will fail due to duplicate key exception

2. Values are distributed across different buckets using linear probing

3. Only one key-value pair will be stored due to overwriting

4. Multiple values are stored in same bucket via chaining

Task 17:
Write a code for binary search tree for expected output

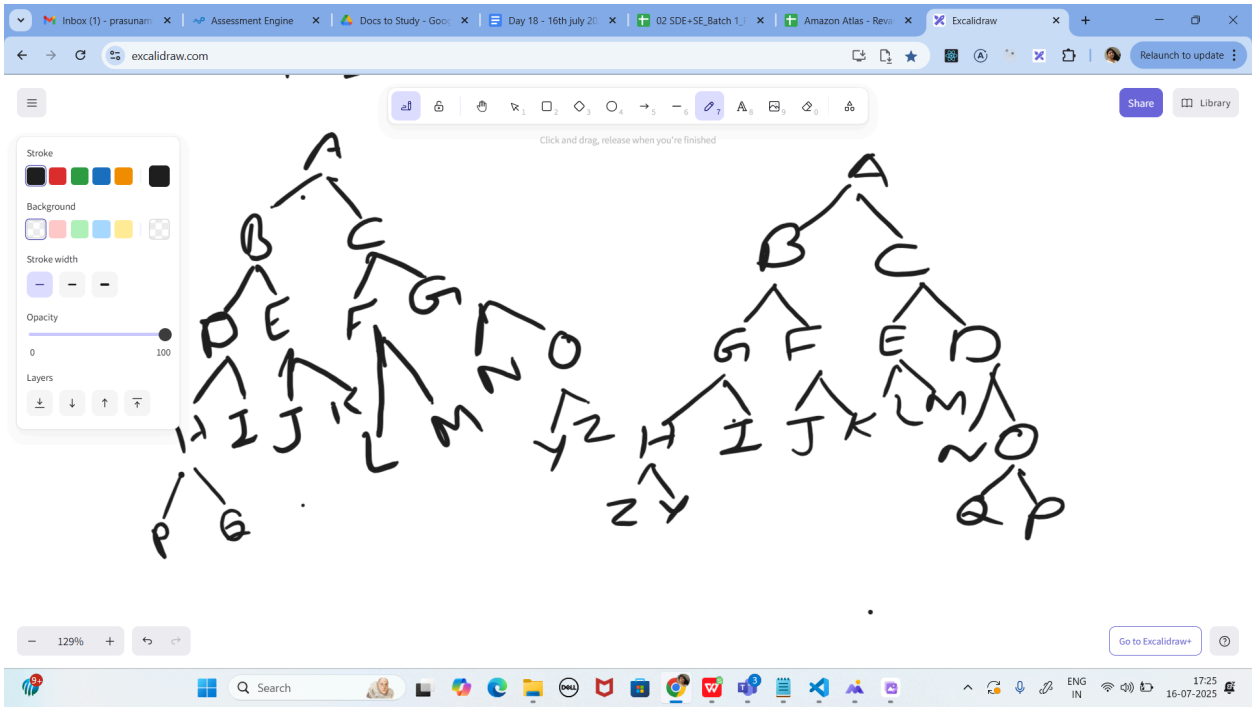
```java
import java.util.*;

class Node {  12 usages
    int key;  2 usages
    Node left, right;  8 usages

    public Node(int key) {  7 usages
        this.key = key;
        left = right = null;
    }
}

class BinaryTreeCornerNodes {
    Node root;  8 usages

    void printCorner(Node root) {  1 usage
        if (root == null) return;

        Queue<Node> q = new LinkedList<>();
        q.add(root);

        while (!q.isEmpty()) {
            int n = q.size();

            for (int i = 0; i < n; i++) {
                Node temp = q.poll();
```

```
            // Print the first and last node of the current level
            if (i == 0 || i == n - 1) {
                System.out.print(temp.key + "  ");
            }

            if (temp.left != null)
                q.add(temp.left);
            if (temp.right != null)
                q.add(temp.right);
        }
    }
}

public static void main(String[] args) {
    BinaryTreeCornerNodes tree = new BinaryTreeCornerNodes();

    // Build the tree
    tree.root = new Node( key: 11);
    tree.root.left = new Node( key: 22);
    tree.root.right = new Node( key: 33);
    tree.root.left.left = new Node( key: 44);
    tree.root.left.right = new Node( key: 55);
    tree.root.right.left = new Node( key: 66);
    tree.root.right.right = new Node( key: 77);
```

```
    public static void main(String[] args) {
        BinaryTreeCornerNodes tree = new BinaryTreeCornerNodes();

        // Build the tree
        tree.root = new Node( key: 11);
        tree.root.left = new Node( key: 22);
        tree.root.right = new Node( key: 33);
        tree.root.left.left = new Node( key: 44);
        tree.root.left.right = new Node( key: 55);
        tree.root.right.left = new Node( key: 66);
        tree.root.right.right = new Node( key: 77);

        // Print corner nodes at each level
        tree.printCorner(tree.root);
    }
}
```

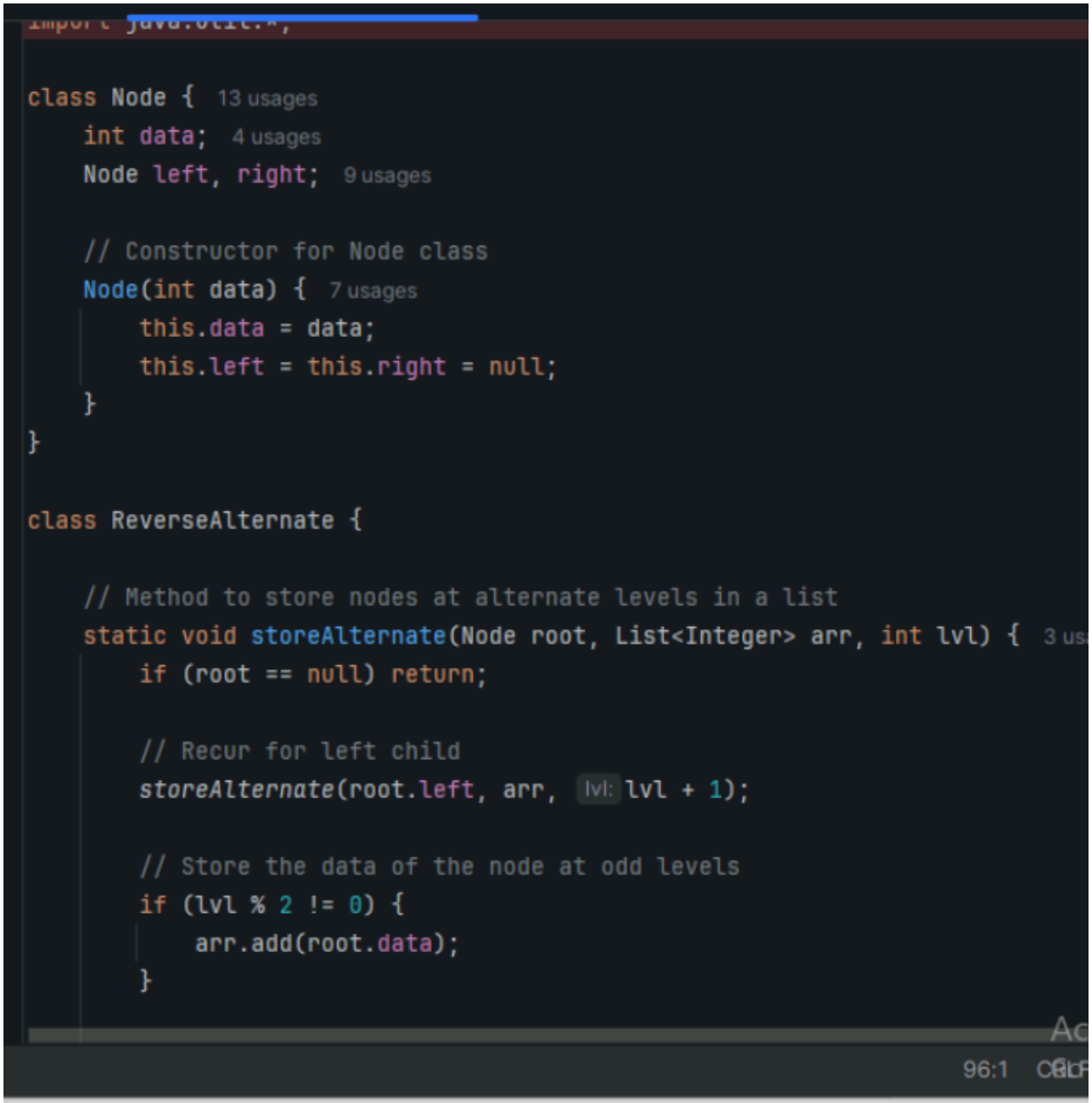
```
"C:\Program Files\Java\jdk-17\bin\java.exe"
11  22  33  44  77
Process finished with exit code 0
```

Task 18:

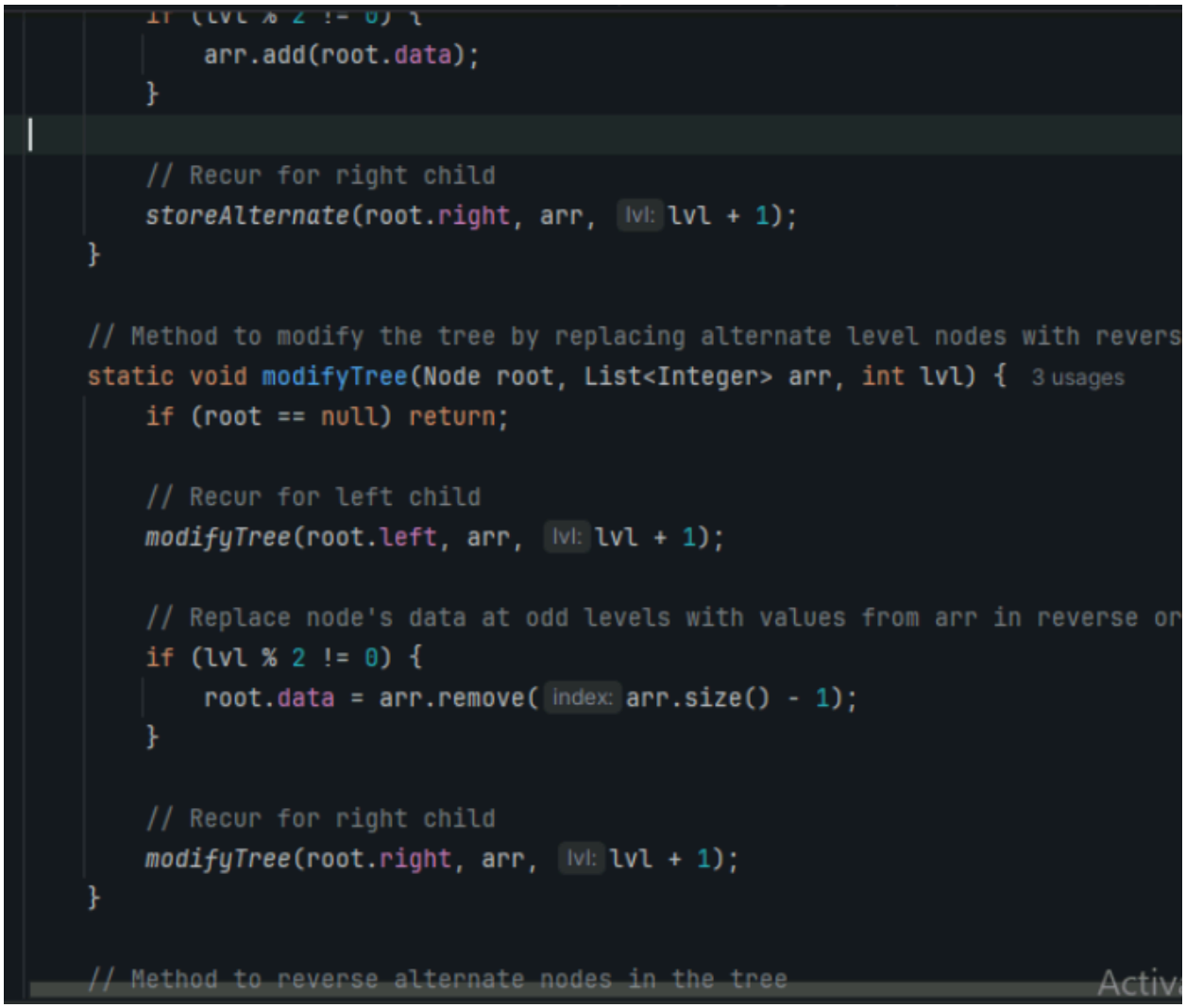



Print reverse order for alternative levels..

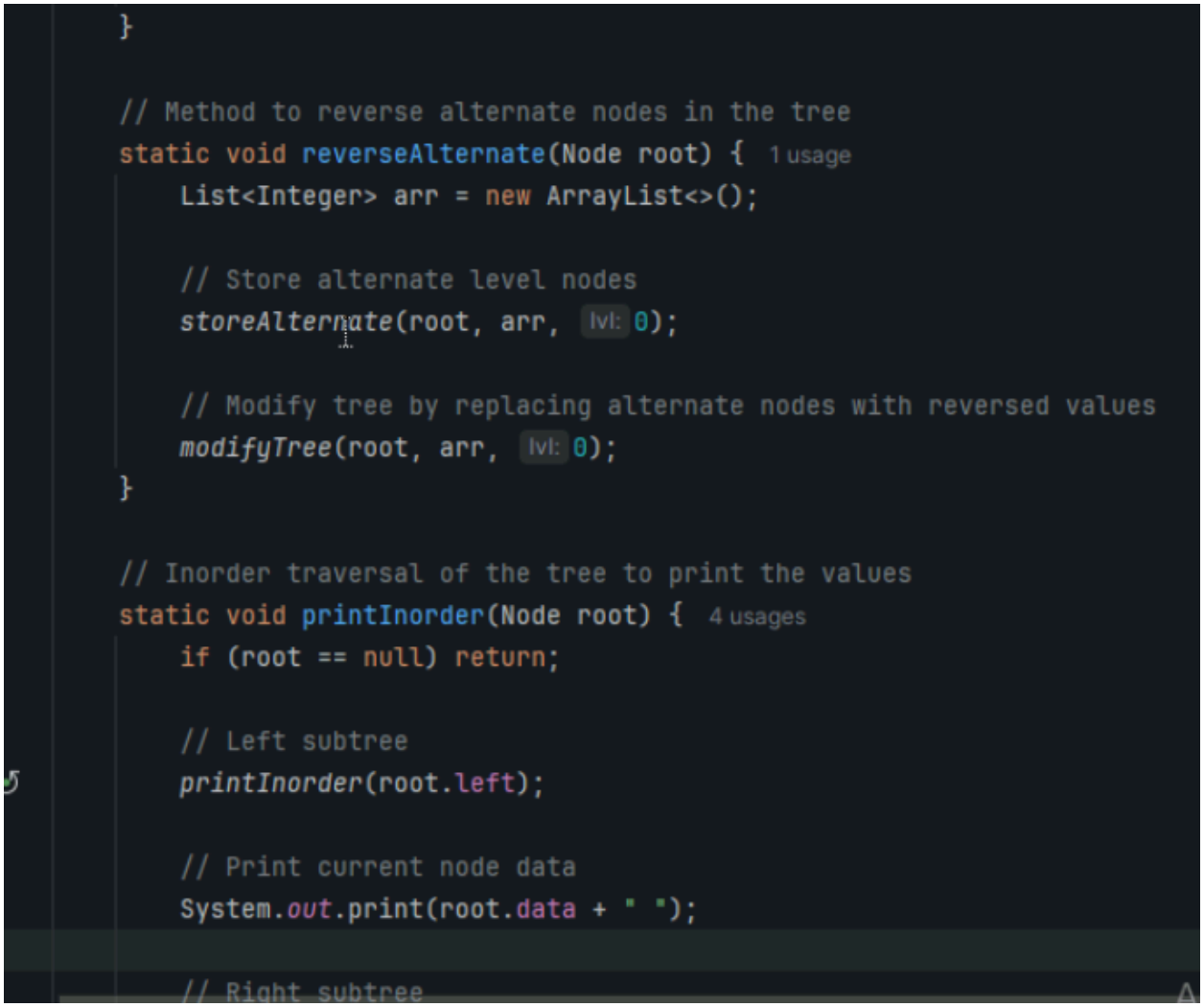
17.26 to 17.36



```
import java.util.*;

class Node {  13 usages
    int data;  4 usages
    Node left, right;  9 usages

    // Constructor for Node class
    Node(int data) {  7 usages
        this.data = data;
        this.left = this.right = null;
    }
}

class ReverseAlternate {

    // Method to store nodes at alternate levels in a list
    static void storeAlternate(Node root, List<Integer> arr, int lvl) {  3 us
        if (root == null) return;

        // Recur for left child
        storeAlternate(root.left, arr,  lvl: lvl + 1);

        // Store the data of the node at odd levels
        if (lvl % 2 != 0) {
            arr.add(root.data);
        }
```



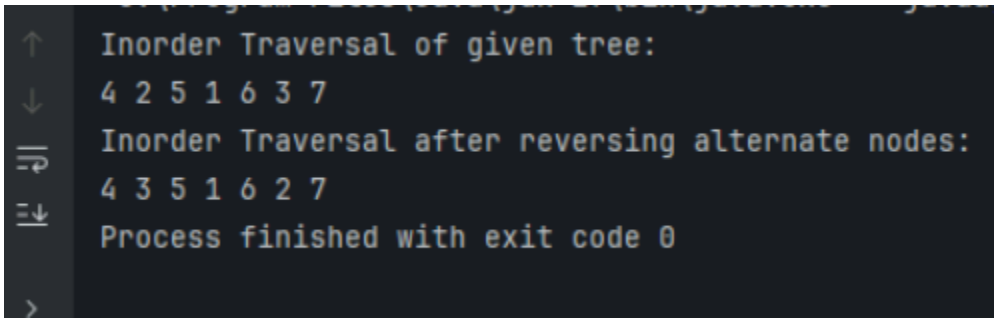
```
        if (lvl % 2 != 0) {
            arr.add(root.data);
        }

        // Recur for right child
        storeAlternate(root.right, arr, lvl: lvl + 1);
    }

    // Method to modify the tree by replacing alternate level nodes with revers
    static void modifyTree(Node root, List<Integer> arr, int lvl) {  3 usages
        if (root == null) return;

        // Recur for left child
        modifyTree(root.left, arr, lvl: lvl + 1);

        // Replace node's data at odd levels with values from arr in reverse or
        if (lvl % 2 != 0) {
            root.data = arr.remove( index: arr.size() - 1);
        }

        // Recur for right child
        modifyTree(root.right, arr, lvl: lvl + 1);
    }

    // Method to reverse alternate nodes in the tree
```



```
    }

    // Method to reverse alternate nodes in the tree
    static void reverseAlternate(Node root) {  1 usage
        List<Integer> arr = new ArrayList<>();

        // Store alternate level nodes
        storeAlternate(root, arr,  lvl: 0);

        // Modify tree by replacing alternate nodes with reversed values
        modifyTree(root, arr,  lvl: 0);
    }

    // Inorder traversal of the tree to print the values
    static void printInorder(Node root) {  4 usages
        if (root == null) return;

        // Left subtree
        printInorder(root.left);

        // Print current node data
        System.out.print(root.data + " ");

        // Right subtree
```

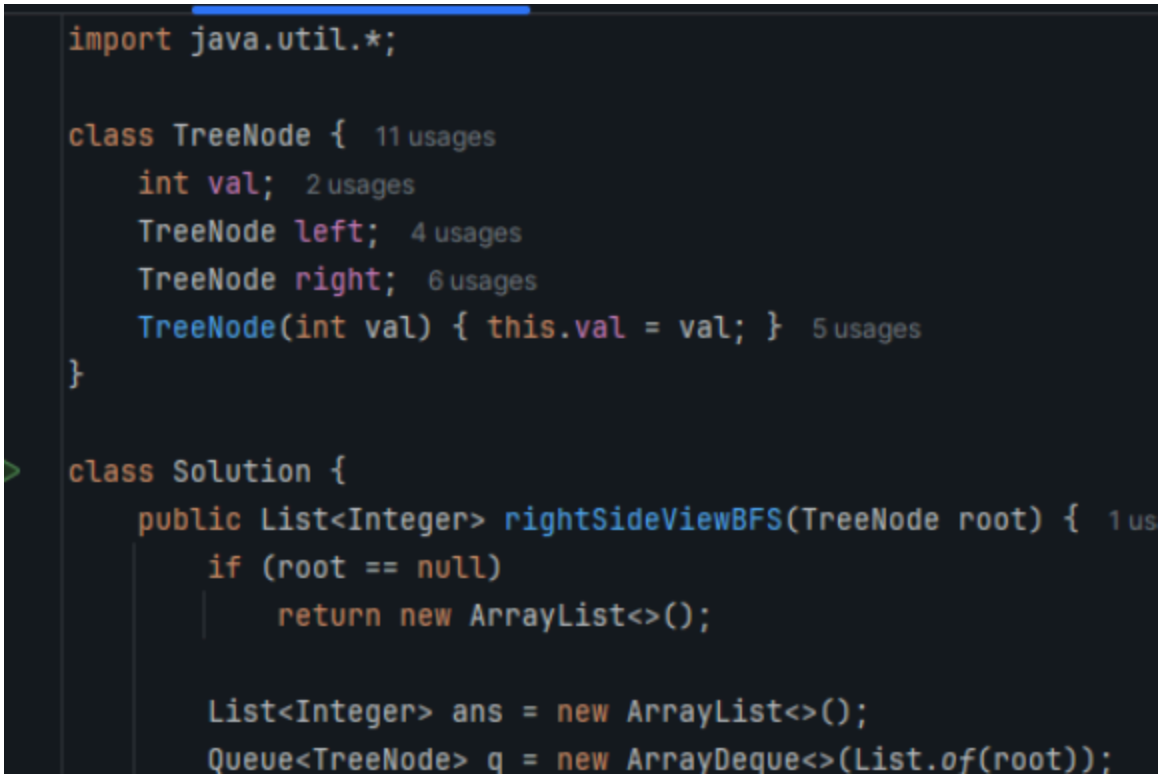
```java
static void printInorder(Node root) {  4 usages
    if (root == null) return;

    // Left subtree
    printInorder(root.left);

    // Print current node data
    System.out.print(root.data + " ");

    // Right subtree
    printInorder(root.right);
}

public static void main(String[] args) {
    // Create the binary tree
    Node root = new Node( data: 1);
    root.left = new Node( data: 2);
    root.right = new Node( data: 3);
    root.left.left = new Node( data: 4);
    root.left.right = new Node( data: 5);
    root.right.left = new Node( data: 6);
    root.right.right = new Node( data: 7);

    // Print inorder traversal before modification
    System.out.println("Inorder Traversal of given tree:");
```

```
        Node root = new Node( data: 1);
        root.left = new Node( data: 2);
        root.right = new Node( data: 3);
        root.left.left = new Node( data: 4);
        root.left.right = new Node( data: 5);
        root.right.left = new Node( data: 6);
        root.right.right = new Node( data: 7);

        // Print inorder traversal before modification
        System.out.println("Inorder Traversal of given tree:");
        printInorder(root);
        System.out.println(); // For formatting

        // Reverse alternate nodes in the binary tree
        reverseAlternate(root);

        // Print inorder traversal after modification
        System.out.println("Inorder Traversal after reversing alternate nodes:");
        printInorder(root);
    }
}
```

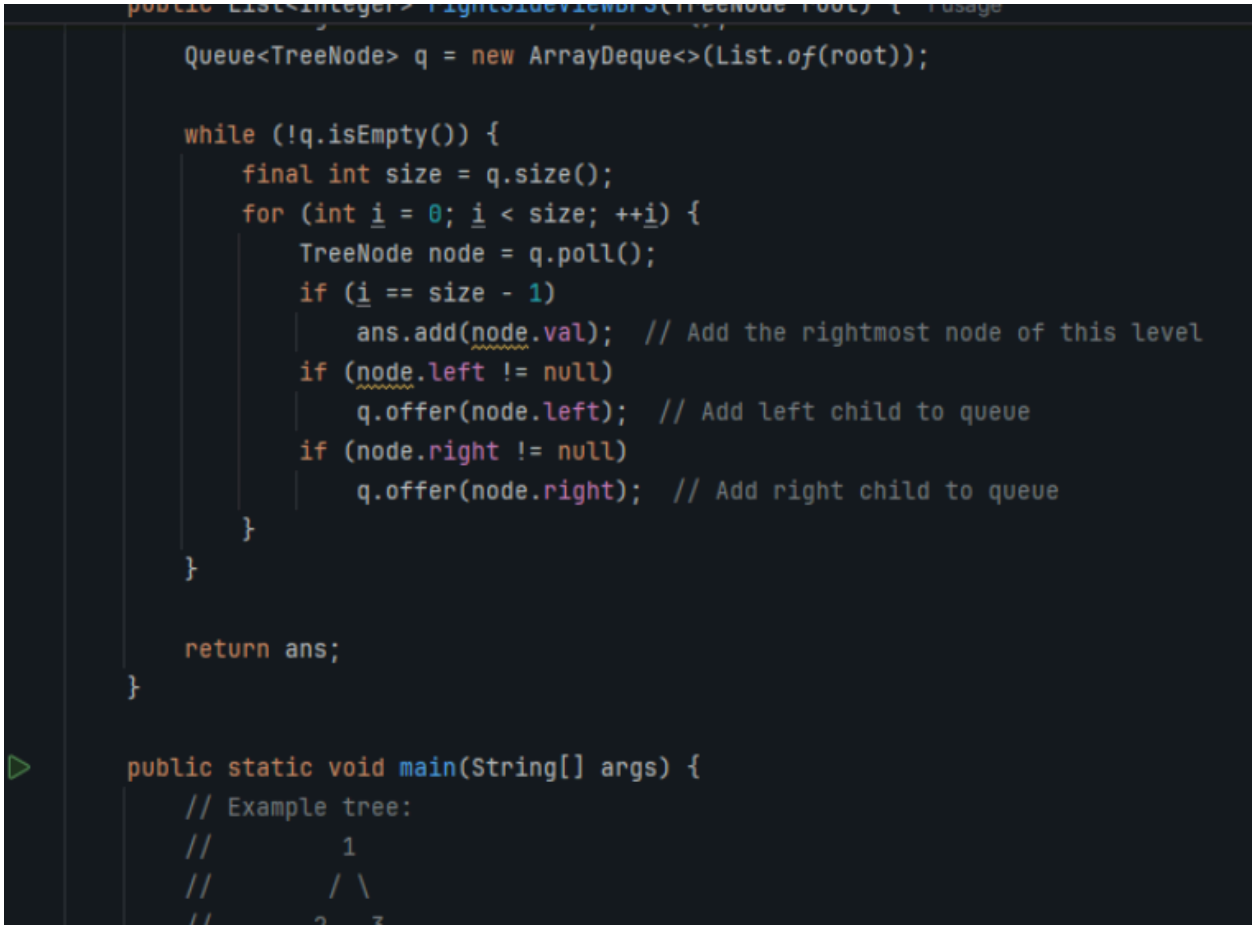



tASK 19
https://leetcode.com/problems/binary-tree-right-side-view/description/

Plz solve this Problem statement



```java
import java.util.*;

class TreeNode {  11 usages
    int val;  2 usages
    TreeNode left;  4 usages
    TreeNode right;  6 usages
    TreeNode(int val) { this.val = val; }  5 usages
}

class Solution {
    public List<Integer> rightSideViewBFS(TreeNode root) {  1 us
        if (root == null)
            return new ArrayList<>();

        List<Integer> ans = new ArrayList<>();
        Queue<TreeNode> q = new ArrayDeque<>(List.of(root));
```



```java
        Queue<TreeNode> q = new ArrayDeque<>(List.of(root));

        while (!q.isEmpty()) {
            final int size = q.size();
            for (int i = 0; i < size; ++i) {
                TreeNode node = q.poll();
                if (i == size - 1)
                    ans.add(node.val);  // Add the rightmost node of this level
                if (node.left != null)
                    q.offer(node.left);  // Add left child to queue
                if (node.right != null)
                    q.offer(node.right);  // Add right child to queue
            }
        }

        return ans;
    }

    public static void main(String[] args) {
        // Example tree:
        //         1
        //        / \
```

```java
    public static void main(String[] args) {
        // Example tree:
        //         1
        //        / \
        //       2   3
        //        \   \
        //         5   4

        TreeNode root = new TreeNode( val: 1);
        root.left = new TreeNode( val: 2);
        root.right = new TreeNode( val: 3);
        root.left.right = new TreeNode( val: 5);
        root.right.right = new TreeNode( val: 4);

        Solution solution = new Solution();
        List<Integer> result = solution.rightSideViewBFS(root);

        System.out.println(result);  // Expected output: [1, 3, 4]
    }
}
```

```
"C:\Program Files\Java\jdk-17\bin\java.exe" "
[1, 3, 4]

Process finished with exit code 0
```